

Automated Segmentation of the Lateral Ventricle Based on Graph Cuts Algorithm and Morphological Operations

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Abstract: Enlargement of the lateral ventricles have been identified as a surrogate marker of neurological disorders. Quantitative measure of the lateral ventricle from MRI would enable earlier and more accurate clinical diagnosis in monitoring disease progression. Even though it requires an automated or semi-automated segmentation method for objective quantification, it is difficult to define lateral ventricles due to insufficient contrast and brightness of structural imaging. In this study, we proposed a fully automated lateral ventricle segmentation method based on a graph cuts algorithm combined with atlas-based segmentation and connected component labeling. Initially, initial seeds for graph cuts were defined by atlas-based segmentation (ATS). They were adjusted by partial volume images in order to provide accurate a priori information on graph cuts. A graph cuts algorithm is to find a global minimum of energy with minimum cut/maximum flow algorithm function on graph. In addition, connected component labeling used to remove false ventricle regions. The proposed method was validated with the well-known tools using the dice similarity index, recall and precision values. The proposed method was significantly higher dice similarity index (0.860 ± 0.036 , $p < 0.001$) and recall (0.833 ± 0.037 , $p < 0.001$) compared with other tools. Therefore, the proposed method yielded a robust and reliable segmentation result.

Key words: Magnetic Resonance Imaging, Graph cuts algorithm, Atlas-based segmentation, Morphological Operation, Partial Volume Estimation

I. Introduction

The lateral ventricle described frequently in several neurological disorders is the largest ventricular system of the brain, in which choroid plexus produces cerebrospinal fluid (CSF). If a generation of CSF is more than reabsorption or its circulation is blocked, the lateral ventricle would be enlarged. An atrophy of adjacent structure might allow expanding the ventricle. An enlargement of the lateral ventricle has been identified as a surrogate marker of neurological disorders such as Alzheimer's disease, schizophrenia, bipolar disorder and mild cognitive impairment, as well as movement disorder such as Parkinson's

disease [1-4]. Since magnetic resonance image (MRI) has been used to diagnose a variety of diseases as a bio-maker, quantitative volume measure of lateral ventricle would enable a noninvasive diagnosis in monitoring disease progression [5,6]. However, various factors render ventricle segmentation difficult despite the high contrast between tissue and CSF. First, partial volume effects can impact segmentation on MRI with limited resolution [7]. Moreover, the temporal horns and occipital poles of the ventricles can be disconnected from the main body, which affects appearance-based methods and region-growing techniques. Finally, the choroid plexus appears with similar intensities to gray matter, which prevents the use of simple threshold-based techniques.

Segmentation of subcortical structures from MR brain scans is a crucial process for volume assessment or shape analysis. Therefore, an accurate and objective methodology is needed for quantitative

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volume measure. However, due to the uncertain boundary with similar intensity to adjacent structures, simple segmentation method like thresholding might generate poor result [8]. Meanwhile, the manual segmentation depending on complete anatomical knowledge is very time consuming and labor intensive. Also, it suffers from inter- and intra-rater variability [4,9].

Several semi-automated methods such as active contours, region growing have been proposed to minimize a user intervention [10,11]. Even though they reduced a certain amount of effort to delineate a region of interest, it was inappropriate to a large study because of manual determination of seed or initial point. On the other hand, several automated methods such as atlas-based segmentation, SPM-ALVIN (<https://sites.google.com/site/mrilateralventricle/>) and FSL-FIRST (<https://fsl.fmrib.ox.ac.uk/fsl/fslwiki/FIRST>) showed a relatively good result for lateral ventricle segmentation [12-14]. However, due to the high variability in shape of lateral ventricle depending on neurodegenerative diseases such as Alzheimer's disease or Parkinson's disease, single automated approach would be difficult to define lateral ventricle exactly.

Recently, hybrid approaches which combine the advantage of diverse segmentation methods have been proposed to overcome several obstacles described above [9,15,16]. Coupe et al. proposed a patch-based method using expert manual delineations as priors [15]. It applied to a lateral ventricle segmentation of patients with Alzheimer's disease, but its robustness and a certain effect of several parameters such as the size of patch and search volume size were not validated. Hu et al. proposed a fully automatic segmentation method combined both level-set and appearance-based modeling techniques [9]. However, it might suffer from mathematical problems and determining control points. Since the segmentation result of single label propagation might be erroneous, Heckemann et al. introduced multiple propagated segmentation of a single-target image as classifiers and combined them using decision fusion rules [16].

In this study, we proposed a fully automated method for segmentation of the lateral ventricle using graph

cuts combined with an atlas-based segmentation and morphological operation. An atlas-based segmentation might generate a biased result due to the risk of systematic error in anatomical labeling, but it can present a priori information as an input data term for graph cuts algorithm [17]. It was applied to healthy normal subjects that had manually defined lateral ventricle regions, and validated using the dice similarity index, recall and precision values. It was also compared with well-known automated segmentation packages such as FSL-FIRST and SPM-ALVIN.

II. Materials and Methods

1. Subjects

Three-dimensional T1-weighted MR images for this study were IXI datasets obtained from Imperial College of Science Technology & Medicine (<http://www.brain-development.org>). Acquisition parameters on the scanner (1.5 Tesla GE Signa Echospeed) had TE = 4.2 ms (fat and water in phase), TR = 15.5 ms, NEX = 1, time of inversion = 450 ms, flip angle 20°, to obtain 124 slices of 1.5 mm thickness with a field of view of 18 × 24 cm with a 192 × 256 matrix. It covered the whole brain with voxel sizes of 0.9375 × 0.9375 × 1.5 mm [18]. Images were performed intensity nonuniformity correction and re-orientation, and re-sliced to create isotropic voxels using windowed sinc interpolation in order to preserve the native resolution and to reduce partial volume artifacts and considerable errors in registration [19-21]. Twenty healthy normal subjects consisting of 10 women (31.6 ± 9.0 years) and 10 men (31.5 ± 9.5) had no neurological, medical, or psychiatric condition and normal MRI studies as determined by two experienced neuro-radiologists. Gold standard used for validation of lateral ventricle segmentation was defined by combining main body, anterior and posterior horns among the 49 areas defined by hand in the process of creating 3D maximum probability atlas [18].

2. Pre-processing

At first, atlas-based segmentation was applied to define lateral ventricle region which would be used a priori information for the following graph cuts

algorithm. A neurologist manually drew lateral ventricle region on the International Consortium for Brain Mapping (ICBM) 152 symmetric template, which was delineated in the axial view while the other views at the same position were displayed simultaneously. It would be used as the target atlas to define the lateral ventricle region from individual data using nonlinear registration.

The following pipeline image processing steps were applied for further analysis, as described in detail elsewhere [3,19,20,22-24]. At first, the native MRI data of all subjects were registered into the template using a linear transformation and corrected for intensity non-uniformity artifacts [20]. And then, an artificial neural network classifier was applied to identify gray matter (GM), white matter (WM) and CSF [24]. Finally, partial volume levels, MRI intensity-mixing at the tissue interfaces due to the finite resolution of the imaging device, were estimated and corrected using a trimmed minimum covariance determinant method [24,25]. Estimating the fractional amount of each tissue type within each voxel improved the accuracy of a priori information for foreground and background voxels in graph cuts algorithm which would be explained later (See ‘Graph cuts approach’ section). A hierarchical multi-scale non-linear fitting algorithm (ANIMAL, Automatic Nonlinear Image Matching and Anatomical Labeling) was then applied to obtain the 3D deformation vector field that maps the template onto the individual brain volume [19]. And then the resulting vector field was used to transform the lateral ventricle atlas [26]. In addition, the component of WM or GM from the transformed lateral ventricle atlas was masked out by CSF-classified map of each subject since the ventricles were filled with CSF. It would be used to exclude the false positive for graph cuts algorithm.

3. Graph cuts algorithm

A graph cuts algorithm was used to find a global minimum of energy function with minimum cut/maximum flow algorithms on graph [17]. A directed weighted graph $G = \langle v, \epsilon \rangle$ consists of a set voxels v of the image and a set of edges ϵ between every neighboring voxels. The graph has two different types of

edge such as N -links (neighborhood links) and T -links (terminal links). An N -link is defined by the connection between neighboring voxels, and a T -link connects between an image voxel and one of two terminals which correspond to foreground and background. The boundary between foreground and background was obtained by finding the minimum cost cut on the graph G , which minimized and energy function which is formulated as:

$$E(f) = \sum_{p \in P} D_p(f_p) + \sum_{\{p,q\} \in N} V_{p,q}(f_p, f_q),$$

where $f = \{f_p | p \in P\}$ denotes a labeling of the image P , and $D_p(\cdot)$ is a unary term derived by cost on the edges of T -links. And $V_{p,q}(\cdot)$ is a pair-wise term derived by discontinuous intensity values between neighboring voxels in the image. The unary term is computed from the histograms of the seeds corresponding to the foreground and background. The foreground seeds consisted of voxels with high probabilities to belong to the CSF, while background seeds were defined by the voxels with high probabilities of the GM and WM (Fig. 1). As a result, the partition of foreground and background was obtained

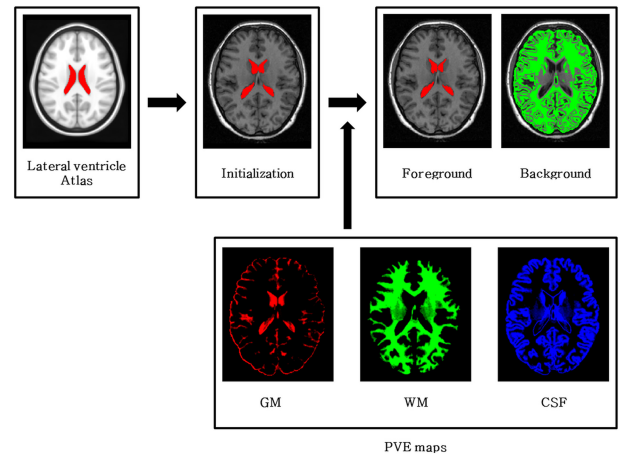


Fig. 1. The processes to define the foreground and background seed automatically. In the first row, each image shows fore- (red) and background (green) defined by the ATS. The second row is the partial volume probability maps that are needed in the processing of seed definition. The probabilistic value of the CSF contributes to generation of the foreground seeds. In a similar way, the probabilistic values of the GM and WM contribute to generation of the background seeds.

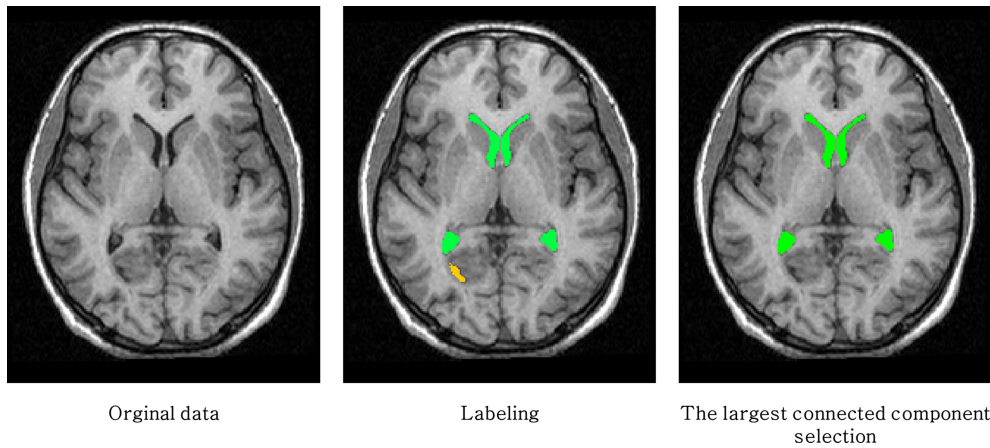


Fig. 2. The result of GC method might include non-ventricle regions (yellow). They were removed by the largest connected component selection, then GCM method showed only ventricle region (green).

by finding the minimum cost cut on the graph G .

4. Connected component labeling

Morphological operations were related to the shape or morphology of features in an image and could be used to remove some imperfections in the segmentation results. Especially, the connected component labeling was used to remove any bias such as non-ventricular region generated from previous atlas-based segmentation (Fig. 2). It generally labels the largest connected component as a lateral ventricle.

5. Validation

Several measures were used for the evaluation of different segmentation results such as atlas-based segmentation, modified one combined with graph cuts approach, its improved version with connected component labeling, and outcomes of well-known tools such as FSL-FIRST and SPM-ALVIN. The dice similarity index is derived from a reliability measure known as the kappa coefficient [27]. We computed the kappa coefficient between the manual definition of lateral ventricle and each segmentation result. This metric measures the similarity of two sets as the ratio of the size of their intersection divided by the sum of areas of the two sets.

$$\kappa(A, B) = \frac{2|A \cap B|}{|A| + |B|}$$

where, A and B are the sets of voxels segmented as

lateral ventricle in manual delineation and each method, respectively. It ranges from 0 for sets that have no common elements to 1 for sets that are identical.

In addition, two well-known validation metrics such as recall and precision were used [28]. Precision is the number of true positive (TP) (i.e. the number of items correctly labeled as belonging to the lateral ventricle) divided by the total number of elements labeled as belonging to the positive class (i.e. the sum of TP and false positive (FP) classes). Recall is defined as the number of TP divided by the total number of elements that actually belong to the positive class (i.e. the sum of TP and false negative (FN) classes).

$$\text{recall} = \frac{A \cap B}{A}, \quad \text{precision} = \frac{A \cap B}{B}$$

A precision score of 1 for a class B means that every item labeled as belonging to class B does indeed belong to class A , but says nothing about the number of items from class B that were not labeled correctly. On the other hand, a recall of 1 means that every item from class B was labeled as belonging to class A , but says nothing about how many other items were incorrectly also labeled as belonging to class B .

Differences between the gold standard and the segmentation results from FSL-FIRST, SPM-ALVIN and the proposed method were examined with a paired t-test using SPSS software package version 18

(SPSS, Inc., Chicago, IL, USA).

III. Results

1. Validation for morphological operation

Fig. 3 shows the effect of morphological operation on the result in terms of the similarity index, recall and precision. In all features, the proposed method (GCM, graph cut method with morphological operation) showed significantly higher (0.860 ± 0.036 , $p < 0.001$) than graph cut method without morphological operation (GC, 0.857 ± 0.037). It meant that the morphological operation ameliorated the segmentation of lateral ventricle.

2. Evaluation with the conventional methods

Fig. 4 shows the bar graph of each index of different segmentation approaches for manual delineation which are atlas-based approach (ATS), FSL-FIRST, SPM-ALVIN and GCM. Even though GCM exhibited significantly higher similarity index and recall value (0.833 ± 0.037 , $p < 0.001$) than the others, its precision value (0.889 ± 0.042) was significantly lower than ATS (0.913 ± 0.043 , $p < 0.001$). It meant that GCM made a tendency of ATS toward over-segmentation better.

The segmentation error was evaluated using the FP and FN maps in stereotaxic space that showed

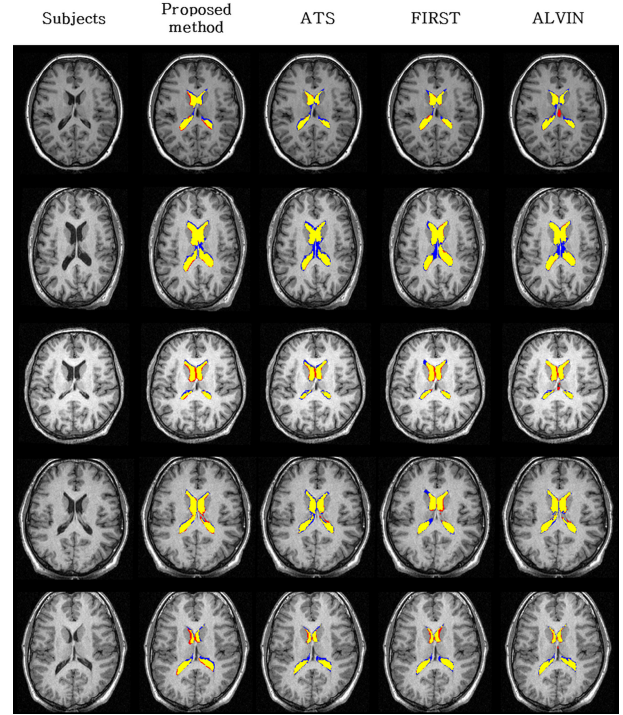


Fig. 5. The example of the segmentation result of each method. The red area denotes the overestimation (false positive, FP) but the blue vice versa (false negative, FN). The overlapped area with the gold standard is shown in yellow color.

condensed spatial distribution of all segmentation results (Fig. 5). The proposed method showed more FP than SPM-ALVIN in the lateral ventricle boundary due to overestimation, while the latter showed some

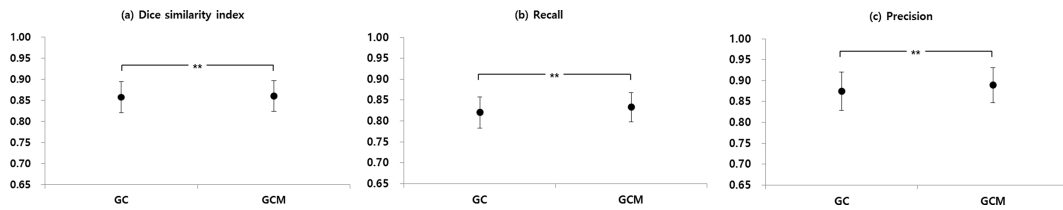


Fig. 3. The bar plot of the Dice similarity index (a), recall (b), and precision values (c) for GC and GCM in reference with gold standard. (** $p < 0.001$, where p indicates the statistical significance).

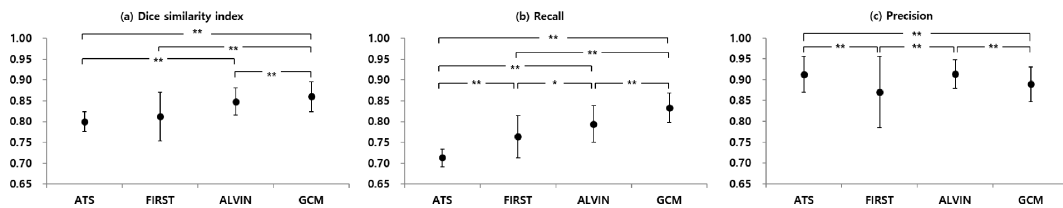


Fig. 4. The bar plot of the Dice similarity index (a), recall (b), and precision values (c) for the different segmentation methods in reference with gold standard. (* $p < 0.05$, ** $p < 0.001$, where p indicates the statistical significance).

underestimation.

Discussion

In this study, we proposed an automated framework for lateral ventricle segmentation method using graph cut algorithm combined with atlas-based segmentation and the connected component selection. The segmentation result of the proposed method was compared with FSL-FIRST and SPM-ALVIN, which are widely employed in lateral ventricle segmentation. The IXI dataset with a manually defined lateral ventricle was used as the gold standard for an objective and righteous validation.

The proposed method demonstrated relatively robust and accurate results. When the proposed method was compared with graph cut approach, the former used largest connected component selection as post-processing. It was to solve the problem for non-ventricular region. Posterior horn regions of lateral ventricle have high variability depending on subjects. To differentiate this region from the result, automated segmentation algorithms needed a lot of anatomical knowledge. H. G. Schnack et al. proposed the box depending on the size of the brain, which could find posterior horn regions empirically [29]. Based on this research, the proposed method with largest connected component selection improved its accuracy significantly by removing overestimated regions.

When we compared three automated segmentations such as ATS, FSL-FIRST and SPM-ALVIN, the latter showed the highest dice similarity index and the former vice versa. ATS relies on the existence of a reference image in which objects of interest have been carefully segmented, manually or with the help of a semi-automatic segmentation method [12,16]. It usually requires the use of image registration in order to align the atlas image to a new image. The main drawback of a single-atlas approach is that if there are significant differences between the atlas and a given test image, then there may not be a good way to map one onto the other. Rather than relying on a single atlas, multi-atlas can be used by propagating manual labels from multiple atlases in a database to a query subject and combining them [29].

The proposed method showed better performance in the dice similarity index and recall than three automated segmentations, but the precision vice versa. It meant that the proposed method was prone to overestimate. In addition, the proposed method showed more FP and less FN than SPM-ALVIN in the boundary of lateral ventricle since the latter included third ventricle region due to the manually drawn its mask. Therefore, the proposed method showed a little overestimated result but could be better approach than the other well-known methods.

In this paper, we proposed an automatic framework for lateral ventricle segmentation method using graph cut algorithm combined with ATS and largest component selection approach and validated its result with healthy subjects. The performance of proposed method was compared with an atlas-based segmentation, FSL-FIRST and SPM-ALVIN. Future studies will modify and apply the proposed method to diagnostic classification of subcortical structures including hippocampus since atrophy pattern of those structures could be used as a biomarker for an early detection of brain diseases.

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